

Focal Dysfunction (Lesions)

A structural lesion such as an infarction, hemorrhage, tumor, or abscess causes local cerebral dysfunction and resultant EEG abnormalities. The EEG is a poorly sensitive and specific test to detect structural brain lesions compared to advanced neuroimaging techniques such as magnetic resonance imaging (MRI) but has a key role in determining the lesion's functional significance (such as epileptogenicity) [1]. Additionally, the EEG can determine physiologic cerebral dysfunction, such as postictal state or hypoperfusion, which are often reversible and without corresponding structural correlate on neuroimaging. The role of the reader is to detect if focal dysfunction is present and estimate its potential severity, etiology (structural or functional), and epileptogenicity.

Basic Approach to Focal Cerebral Dysfunction on EEG

1. detection
2. estimate severity
3. estimate if structural (fixed) or functional (reversible) underlying etiology
4. estimate potential for seizures (epileptogenicity)

Detecting Focal Cerebral Dysfunction

Two key EEG features indicate the presence of focal dysfunction:

1. **Disruption of background activity in the ipsilateral hemisphere:** During wakefulness, there may be asymmetric slowing, amplitude attenuation, or decreased reactivity of the posterior-dominant rhythm (Bancaud's phenomena refers to the persistence of alpha during eye opening on the side of hemispheric dysfunction). Anterior faster frequencies (beta) may also be asymmetrically decreased. During drowsiness/sleep, there may be an asymmetric disruption of sleep architecture. Asymmetries in photic driving response and/or hyperventilation build-up may be elicited during activation procedures.
2. **Emergence of focal slowing (delta or theta waves):** Focal abnormalities indicate local cerebral dysfunction. Focal slowing (theta or delta waves) is the most common abnormality. It may occur sporadically or in abundance and is thought to be caused by the disruption of cortical connections in the cerebral white matter [2].

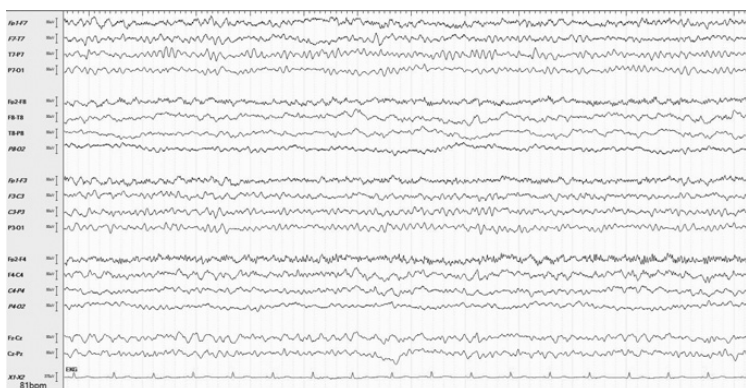


Figure 22.1 Sixty-seven-year-old woman with glioblastoma, right hemispheric dysfunction that is maximal over the temporal region.

Figure 22.1 shows right hemispheric dysfunction worse over the temporal region and characterized by disruption of the normal background and polymorphic slowing.

Estimating Severity

Nonspecific electrographic markers of severity include the abundance of focal slowing in a record, attenuation of amplitude, increased slower frequencies, and the loss of reactivity to external stimulation or state change (usually recorded by the technologist). Milder forms of focal dysfunction may be accentuated during drowsiness. Some sporadic subtle temporal slowing in the elderly is not uncommon during drowsiness and may not have any clinical significance.

Structural (Fixed) versus Physiologic (Reversible) Focal Dysfunction

Reactive, intermittent (sporadic or bursts), or fluctuating slowing may indicate physiological dysfunction such as transient hypoperfusion (vasospasm or hemiplegic migraines) or a postictal state as shown in Chapter 14, Figure 14.14. These abnormalities are reversible and disappear on repeat recordings. There may not be any associated abnormalities on neuroimaging.

Structural (fixed) lesions are typically associated with abundant (often continuous) focal mixed polymorphic theta and delta slowing that does not

react (alter or improve) to external stimulation or state change (arousal) as shown in Chapter 14, Figure 14.15.

These features, however, are neither sensitive nor specific.

Potential Epileptogenicity

Focal intermittent rhythmic (monomorphic) delta activity such as lateralized rhythmic delta activity (LRDA) of which temporal intermittent rhythmic delta activity (TIRDA) is a subtype are considered an epileptogenic abnormality (despite lacking sharp morphology) due to increased association with epileptic seizures as shown in Chapter 15, Figure 15.4.

Focal sporadic (sharps/spikes) or periodic (LPDs) epileptic discharges may also occur. Figure 22.2 shows continuous polymorphic delta slowing and paracentral periodic discharges over the left hemisphere.

A breach effect from an overlying skull defect (such as a craniotomy) may cause enhancement of amplitudes and sharpness that make the underlying cortical activity appear “epileptiform” as shown in Figure 22.3, caution should be exercised when interpreting sharp waves through a breach effect to avoid overcall. The technologist must note the location of skull defects.

Note that dysfunction may occur independently over both hemispheres as shown in Figure 22.4 (multifocal dysfunction). Focal dysfunction may also occur in the setting of diffuse cerebral dysfunction (encephalopathy) [3].

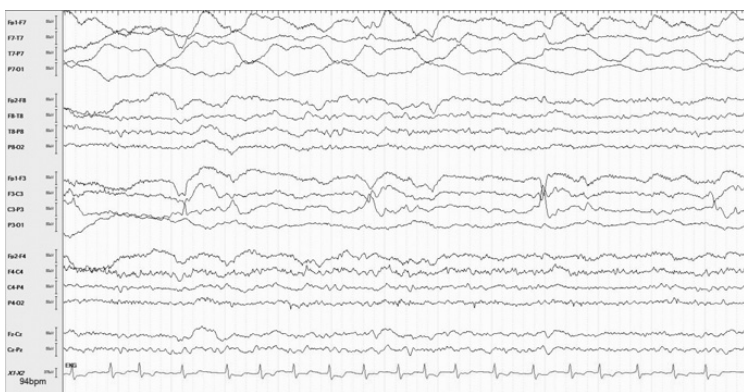


Figure 22.2 Sixty-three-year-old man with a left hemispheric stroke, predominant left hemispheric dysfunction, and cortical irritability. Additionally, there is evidence of a mild encephalopathy.

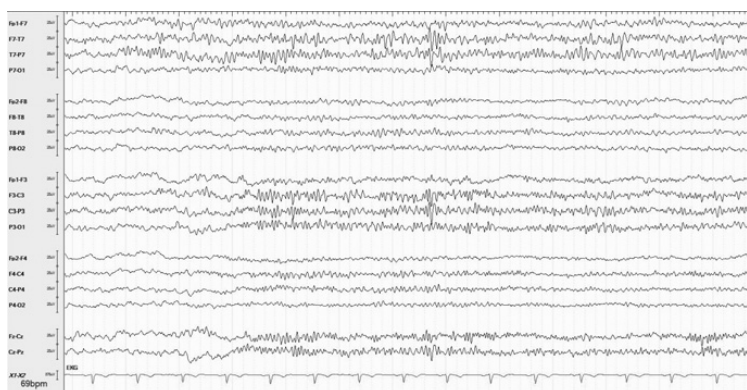


Figure 22.3 Sixty-four-year-old man with a craniotomy for subdural hematoma, left hemispheric breach effect.

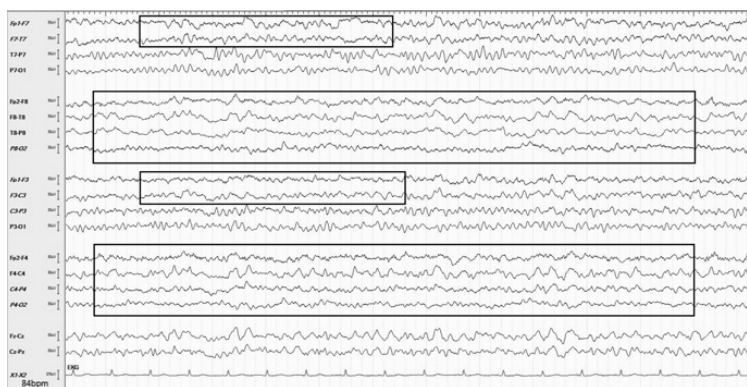


Figure 22.4 Same patient as in Figure 22.1, left frontal and prominent right hemispheric dysfunction.

Chapter Summary

1. The EEG is poorly sensitive and specific to detect lesions compared to neuroimaging; its practical use is to determine the functional consequence of the lesion.

2. Focal dysfunction (physiologic) may occur without an associated neuroimaging abnormality. Postictal states and hypoperfusion are examples of physiologic dysfunction, these are often reversible (disappear on repeat testing).
3. Focal dysfunction causes disruption of the ipsilateral hemispheric background architecture (wakefulness and sleep), asymmetric responses on activation procedures, and focal slowing.
4. Severity of the focal dysfunction may be estimated based on the abundance of focal slowing, attenuation of amplitude, and loss of reactivity.
5. Sporadic, intermittent, or fluctuating focal slowing that is reactive to external stimulation or endogenous state changes (such as arousal) may indicate physiological dysfunction.
6. Focal intermittent rhythmic (monomorphic) delta activity such as lateralized rhythmic delta activity (LRDA), specifically indicates epileptogenicity. It should be treated like an epileptic discharge despite the lack of a sharpness. Look for epileptic discharges that may accompany focal slowing.
7. Focal slowing may occur in isolation, bilaterally, or in the setting of diffuse cerebral dysfunction.

References

1. Noh BH, Berg AT, Nordli DR Jr. Concordance of MRI lesions and EEG focal slowing in children with nonsyndromic epilepsy. *Epilepsia*. 2013 Mar;**54**(3):455–60.
2. Britton JW, Frey LC, Hopp JL, et al. *Electroencephalography (EEG): an introductory text and atlas of normal and abnormal findings in adults, children, and infants*. American Epilepsy Society, Chicago; 2016.
3. Gaspard N, Manganas L, Rampal N, Petroff OA, Hirsch LJ. Similarity of lateralized rhythmic delta activity to periodic lateralized epileptiform discharges in critically ill patients. *JAMA Neurology*. 2013 Oct 1;**70**(10):1288–95.